

Core Modules & Functional Features

1. Health Care Facility (HCF) Portal

Provides hospitals and clinics with self-service tools for compliance and dispatch.

- **Dashboard:** Shows stats on waste bags generated, pending collection, and already dispatched.
- **Self-Scanning Dispatch Protocol:**
 - Restricted self-scanning triggers automatically for bedded facilities (e.g. >30 beds) to ensure bags are scanned and weighed by hospital staff.
 - Generates audit trails during dispatch to driver routes.
- **Facility Certificates Section:** Allows HCFs to view, print, or download official CBWTF Treatment Certificates specific to their facility.

2. Driver mobile web app

Web-based workflow tailored for drivers picking up waste.

- **Route Management:** Drivers select their vehicle, check in with GPS geolocation coordinates, and start their daily collection trip.
- **QR Code Bag Pickup:** Scans bag labels, validates ownership, and checks status.
- **Live Bluetooth Scale Integration:** Pairs with digital weighing scales over Web Bluetooth to auto-capture exact bag weights (metric/imperial parsed to kg).
- **Offline Sync Queue:** Caches collection scans and GPS points in browser memory, auto-synchronizing to Supabase when returning to mobile coverage.
- **Manifests:** Generates collection manifests (Yellow, Red, Blue, White bag counts and weights) signed off on pickup.

3. Plant Operations Module

Manages gate arrivals, batching, and waste processing.

- **Gate Scan:** Verifies vehicle arrivals and scans incoming bags at the plant gate to match against driver manifests.
- **Reconciliation Engine:** Compares bag counts and weights scanned at the HCF against those scanned at the plant gate. Automatically triggers discrepancy alerts for missing bags or weight variations.
- **Batch Processing:** Groups reconciled bags into batches with unique identifiers, preparing them for final treatment.
- **Treatment & Disposal:** Mark batches as treated via compliant methods (Autoclave, Incineration, Microwave, Chemical, Hydroclave) and print/save legal disposal certificates.

4. Regulatory & Administrative Control Panel

High-level command center for CBWTF admins and pollution control regulators.

- **Compliance Dashboard:** Real-time metrics, charts (pie/bar) showing category-wise waste volume, and active driver maps.
- **HCF Registry:** Manage bed capacity, district configurations, facility codes (HCF0001), and hospital

types (bedded/non-bedded).

- **User Management:** Create and manage profiles with strict role scopes (Plant Head, Manager, Driver, HCF Staff, Regulatory).

- **Discrepancy Resolution:** Worklist of flagged weight mismatches or missing bags with operators entering resolution comments to close cases.

- **Form IV Reports Engine:** Generates monthly and annual regulatory compliance reports, downloadable in printable formats.

- **Immutable Audit Trail:** Log of all scan events, GPS check-ins, and operator actions, secured by database row-level security.
